

Exception Handling

Exception Handling in Java is a mechanism used to handle runtime errors so that the normal flow of the program can continue.

An **Exception** is an unwanted event that occurs during program execution, such as dividing by zero, accessing an invalid array index, or opening a non-existing file.

Why Exception Handling?

- Prevents program termination due to errors.
- Maintains normal program flow.
- Improves program reliability and readability.

Without Exception Handling

```
public class Main {  
    public static void main(String[] args) {  
        int a = 10;  
        int b = 0;  
        System.out.println(a / b);  
        System.out.println("Program Continues...");  
    }  
}
```

Compiler gives no error.

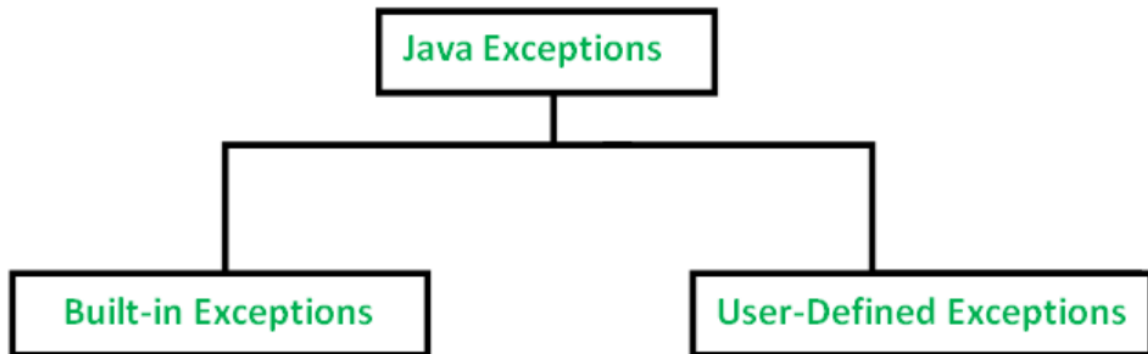
Program crashes at runtime.

"Program Continues..." is not printed.

With Exception Handling

```
public class Main {  
    public static void main(String[] args) {  
        int a = 10;  
        int b = 0;  
        try {  
            System.out.println(a / b);  
        } catch (ArithmeticException e) {  
            System.out.println("Cannot divide by zero.");  
        }  
    }  
}
```

```
System.out.println("Program Continues...");
}
}
Cannot divide by zero
Program Continues...
```



Types of Exceptions

Checked Exceptions

A checked exception is an exception that is checked by the Java compiler

```
// Checked Exception: FileNotFoundException
import java.io.*;

class CheckedExceptionDemo {
    public static void main(String[] args) {
        try {
            FileReader file = new FileReader("data.txt");
        } catch (FileNotFoundException e) {
            System.out.println("File not found.");
        }
    }
}
```

```
// ClassNotFoundException
public class Demo {
    public static void main(String[] args) {
```

```
try {
    Class.forName("Employee");
    System.out.println("Class Found");
} catch (ClassNotFoundException e) {
    System.out.println("Class Not Found");
}
}
```

Unchecked Exceptions

Checked at runtime

Examples:

- ArithmeticException
- NullPointerException
- ArrayIndexOutOfBoundsException

// ArithmeticException (Divide by Zero)

```
class UncheckedExceptionDemo {
    public static void main(String[] args) {
        int a = 10;
        int b = 0;

        try {
            int result = a / b;
        } catch (ArithmeticException e) {
            System.out.println("Cannot divide by zero.");
        }
    }
}
```

// NullPointerException

```
public class Demo {
    public static void main(String[] args) {
```

```
String name = null;
System.out.println(name.length());
}
}
```

// ArrayIndexOutOfBoundsException

```
public class Demo {
    public static void main(String[] args) {
        int a[] = {10, 20, 30};
        System.out.println(a[5]);
    }
}
```

// StringIndexOutOfBoundsException

```
public class Demo {
    public static void main(String[] args) {
        String s = "Java";
        System.out.println(s.charAt(10));
    }
}
```

NumberFormatException

//Occurs when a string cannot be converted to a number

```
public class Demo {
    public static void main(String[] args) {
        String s = "ABC";
        int n = Integer.parseInt(s);
        System.out.println(n);
    }
}
```

IllegalArgumentException

//Occurs when a method receives an illegal argument

```
public class Demo {
    public static void main(String[] args) {
        Thread t = new Thread();
    }
}
```

```
t.setPriority(15); // Valid priorities: 1 to 10
}
}
```

NegativeArraySizeException

```
public class Demo {
    public static void main(String[] args) {

        int a[] = new int[-5];
    }
}
```

Exception Handling Keywords

try

Contains code that may generate an exception

```
try {
    int result = 10 / 0;
}
```

catch

Handles the exception

```
try {
    int result = 10 / 0;
}
catch (ArithmeticException e) {
    System.out.println("Exception handled.");
}
```

finally

Always executes whether an exception occurs or not.

Throw: Explicitly throws an exception

```
public class Demo {
    public static void main(String[] args) {
        int age = 15;
        if (age < 18) {
            throw new IllegalArgumentException("Age must be 18 or above");
        }
    }
}
```

```
}  
    System.out.println("Eligible");  
}  
}
```

Exception in thread "main"

java.lang.IllegalArgumentException: Age must be 18 or above

Throws: Declares that this method may throw an exception

```
import java.io.FileNotFoundException;  
  
public class CheckedExceptionDemo {  
    static void checkFile() throws FileNotFoundException {  
        throw new FileNotFoundException("data.txt not found");  
    }  
    public static void main(String[] args) {  
        try {  
            checkFile();  
        } catch (FileNotFoundException e) {  
            System.out.println(e.getMessage());  
        }  
    }  
}
```

data.txt not found

```
import java.io.IOException;  
  
public class Demo {  
    // throws: Declares that this method may throw IOException  
    static void checkAge(int age) throws IOException {  
        if (age < 18) {  
            // throw: Explicitly throws an exception  
            throw new IOException("Age must be 18 or above.");  
        }  
    }  
}
```

```
System.out.println("Eligible to vote.");
}
public static void main(String[] args) {
    try {
        checkAge(15);
    }
    catch (IOException e) {
        System.out.println(e.getMessage());
    }
    finally {
        System.out.println("Finally block executed.");
    }
    System.out.println("Program Ends.");
}
}
```

Age must be 18 or above.

Finally block executed.

Program Ends.

Multiple Catch Blocks

```
public class MultipleCatchDemo {
    public static void main(String[] args) {
        try {
            int a = 10;
            int b = 0;
            int result = a / b; // ArithmeticException
            int arr[] = {1, 2, 3};
            System.out.println(arr[5]); // ArrayIndexOutOfBoundsException
        } catch (ArithmeticException e) {
            System.out.println("Cannot divide by zero.");
        } catch (ArrayIndexOutOfBoundsException e) {
            System.out.println("Array index is invalid.");
        } catch (Exception e) {
```

```
        System.out.println("Some other exception occurred.");
    }
    System.out.println("Program continues...");
}
}
```

Cannot divide by zero.

Program continues...

User-Defined Exception

A user-defined exception (also called a **custom exception**) is an exception created by the programmer to handle application-specific errors.

Example 1: Invalid Age

```
class InvalidAgeException extends Exception {
    InvalidAgeException(String message) {
        super(message);
    }
}

public class Main {
    static void checkAge(int age) throws InvalidAgeException {
        if(age < 18)
            throw new InvalidAgeException("Not Eligible to Vote");
        else
            System.out.println("Eligible to Vote");
    }

    public static void main(String args[]) {
        try {
            checkAge(16);
        }
        catch(InvalidAgeException e) {
            System.out.println(e.getMessage());
        }
    }
}
```



```
}
```

Not Eligible to Vote

Example 2: Insufficient Balance

```
class InsufficientBalanceException extends Exception {  
    InsufficientBalanceException(String message) {  
        super(message);  
    }  
}  
  
public class Bank {  
    static void withdraw(int balance, int amount)  
        throws InsufficientBalanceException {  
        if(amount > balance)  
            throw new InsufficientBalanceException("Insufficient Balance");  
        else  
            System.out.println("Amount Withdrawn");  
    }  
  
    public static void main(String args[]) {  
        try {  
            withdraw(5000,6000);  
        }  
        catch(InsufficientBalanceException e) {  
            System.out.println(e.getMessage());  
        }  
    }  
}
```

Insufficient Balance

Example 3: Invalid Password

```
class InvalidPasswordException extends Exception {  
    InvalidPasswordException(String msg) {  
        super(msg);  
    }  
}
```



```
}  
public class Login {  
    public static void main(String args[]) {  
        String password = "abc123";  
        try {  
            if(!password.equals("admin123"))  
                throw new InvalidPasswordException("Wrong Password");  
            System.out.println("Login Successful");  
        }  
        catch(InvalidPasswordException e) {  
            System.out.println(e.getMessage());  
        }  
    }  
}
```